

SIMATS ENGINEERING

Department of Computer Science and Engineering



CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison

Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	MUKILAN.R
Register Number	192421043
Class / Slot	Slot B
Capstone Title	SkillHire Cloud - Resume-to-Job Matcher Using NLP Skill Gap Intelligence
Capstone Product Type	Web app / Android app / iOS app / Hybrid app
CO1 Architecture Reference	Mention concept map / presentation title / GitHub link if available

Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	Job seekers, recruiters, and platform admins
User roles	Job Seeker, Recruiter, Admin
Main modules	Resume upload, NLP resume parsing, skill-gap analysis, job matching, recommendation engine, recruiter dashboard, admin dashboard, analytics dashboard
Cloud services identified in CO1	IaaS (VMs, networking, storage), PaaS (Spring Boot backend deployment), SaaS (web application for end users), XaaS (AI/NLP APIs, monitoring, analytics)
API/backend need	REST API (Spring Boot) for login, resume upload, parsing, job matching and dashboards
Database need	Relational database (MySQL) to store users, resumes, job listings, skills and match scores
File upload/storage need	Cloud object storage for resume PDF/DOCX files, generated reports and profile images
Analytics/AI/Python module	Python NLP module (spaCy/NLTK) for tokenization, skill extraction and skill-gap scoring

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

My capstone workload type and reason: SkillHire is primarily an AI/Python module combined with an API backend and a database-heavy workload. Resume parsing and skill-gap scoring (spaCy/NLTK) are CPU- and memory-intensive, job matching queries the database frequently, and the REST API handles authentication and dashboard traffic. Starting guidance: 4-8 vCPU, 16-32 GB RAM for the NLP/API tier, with the database run as a separate managed service rather than combined on one small VM.

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Current Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100
Requests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
Requests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Daily Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000
Input	Value	Reason / Assumption
Projected registered users expected	5,500	Projected job seekers (5,000) + recruiters (500) + admins (10) in year 1
Active user percentage	10%	Typical for an early-stage platform; only a fraction log in on a given day
Peak factor	2x	Accounts for traffic spikes during hiring drives / placement season
Actions per user per minute	2	Average of browsing dashboard, uploading a resume, and viewing matches
Calculated RPS	~36.7 RPS	Concurrent = 5500 x 10% x 2 = 1100; Req/min = 1100 x 2 = 2200 RPS = 2200 / 60 = 36.7
Daily active users	1,500	Estimated from typical daily login pattern of the registered user base
Daily requests	30,000	Daily Requests = 1500 daily active users x 20 avg actions/day = 30,000

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
Simple web/API application	1
User authentication and role management	+1
Moderate API traffic above 5 RPS	+1
Database-heavy search or reporting	+1 to +2
Background jobs or notifications	+1
Analytics or dashboards	+2
Python inference inside server	+2 to +4
Total CPU Score	Recommended vCPU
1-2	1-2 vCPU
3-4	2 vCPU
5-6	4 vCPU

7-9		4-8 vCPU or separate analytics/AI service	
10+		Scale out with multiple VMs or managed compute	
CPU Driver Selected		Score	Reason
Base web/API application	1	Every module runs on a Spring Boot REST API base	
Authentication and roles	+1	JWT-based login and role checks for Job Seeker / Recruiter / Admin	
High traffic	+1	~36.7 RPS estimated at peak, above the 5 RPS threshold	
Database/search/reporting	+1	Job search, skill matching and dashboard reporting query MySQL frequently	
Background jobs	+1	Notification emails and periodic re-matching jobs run in the background	
Analytics/dashboard	+2	Recruiter and admin dashboards aggregate match scores and hiring trends	
AI/Python module	+3	NLP resume parsing, tokenization and skill-gap scoring run inside the application tier	
Total CPU Score	10	Sum of all CPU driver scores above (1+1+1+1+1+2+3 = 10)	
Recommended vCPU	4-8 vCPU (separate NLP/AI service)	Score of 10 falls in the 10+ band; split the NLP workload onto its own VM instead of overloading a single general-purpose VM	

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula		
Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer. If database is managed separately, do not include full database memory in the app VM.		
Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2 GB
Web/API service	1-2 GB	2 GB
Small database	2-4 GB	Not included - managed separately (MySQL runs as its own service)
Medium database/cache	4-8 GB	Not included - managed separately (MySQL runs as its own service)
Analytics/reporting	4-8 GB extra	4 GB
AI/Python module	4-16 GB extra	8 GB
25% buffer	Subtotal x 0.25	Subtotal (2+2+4+8=16) x 0.25 = 4 GB
Final RAM	Round up to standard size	16 + 4 = 20 GB -> rounded up to 32 GB standard tier

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
OS and application	20-40 GB	30 GB
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	~20 GB initial (avg record 2 KB x ~5,510 users x ~50 records/user x 1.3 buffer), growing ~10 GB/year
File uploads	Avg file MB x files/day x days x 1.3	~100 GB (avg resume 1 MB x ~500 uploads/day x 200 days x 1.3 buffer, kept in object storage)
Logs	Requests/day x log KB x days / 1024 MB	~10 GB (~30,000 requests/day x 2 KB log entry x 180 days / 1024)
Backup/growth buffer	Subtotal x 30%	Subtotal (30+20+100+10=160) x 30% = 48 GB
Final storage	Round up to standard disk size	160 + 48 = 208 GB -> rounded up to 250 GB block storage, plus a separate ~150 GB object storage bucket for resumes/reports

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	Used during initial development and faculty demo phase
Two-tier design	App VM + managed/separate database	Pilot phase - app VM separated from the managed MySQL database
Three-tier design	Frontend, backend/API and database separated	Production target - frontend, backend/API+NLP and database tiers separated
Containerized backend	Microservices, portable API, CI/CD	Backend REST API and NLP module containerized for portability and CI/CD
Serverless function	Event-based task, notification, small automation	Used for notification emails and scheduled re-matching jobs
Managed database/storage	Reliability, backup and scaling required	MySQL managed service + object storage for resumes, for reliability and backup

Final VM Plan: Three-tier design (App VM + separate NLP/AI service + managed MySQL).

App/API VM: 2 vCPU, 8 GB RAM (Standard B2s equivalent) running Spring Boot + Nginx.

NLP/AI VM: 4 vCPU, 16 GB RAM dedicated to resume parsing, skill extraction and matching, since the AI/Python driver scored highest in the CPU calibration table.

Database: Managed MySQL 8.0, 20 GB initial storage, multi-zone availability, sized independently of the app VM's RAM.

Storage: ~250 GB block storage for OS/app/logs plus a ~150 GB object storage bucket for resumes, reports and images.

Reason: The CPU score of 10 and the AI/Python-heavy NLP module justify separating the NLP service onto its own VM rather than overloading a single general-purpose VM, while the managed database keeps operational risk and backup effort low.

Activity 8: Hypervisor and Virtualization Decision

Approach		Meaning	Best Use
Type 1 hypervisor		Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor		Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization		Cloud provider hides hypervisor details	Most public cloud VM deployments
Container runtime		OS-level process isolation	Lightweight APIs, microservices, fast deployment
Context		Selected Approach	Justification
Local prototype/testing		Type 2 hypervisor	Used on the student's laptop (VirtualBox/VMware Workstation) to build and test the Spring Boot app and NLP scripts before any cloud deployment
Cloud pilot deployment		Cloud-managed virtualization	Deploying the pilot on a public cloud VM hides hypervisor management so the team can focus on the application
Production growth scenario		Cloud-managed virtualization + container runtime	As user load grows, the backend and NLP services move into containers behind an autoscaling group on the provider's Type 1 hypervisor infrastructure
Module suitable for container		Container runtime	The REST API and NLP microservice are stateless and benefit from fast startup, portability and CI/CD via containers (e.g., Docker)

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment	Snapshot	Immediately before each release	Retained 7 days, rollback safety net
Before database schema change	Snapshot	Just before running migration scripts	Retained until migration verified (~3 days)
Before OS/package update	Snapshot	Before applying OS/security patches	Retained 7 days
Before review/demo	Snapshot	A few hours before faculty review/demo	Retained until after the demo
Create testing environment	Clone	When QA needs an isolated environment	Kept for the duration of the test cycle

Create team demo copy	Clone	Before hackathon/team demo sessions	Kept until the demo event is complete
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Technical Tip

A snapshot is useful for rollback. A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning
Capstone Module	VM / Container / Managed Service	Reason
Frontend	Managed Service (static hosting/CDN)	Simple HTML/CSS/JS frontend benefits from a managed static hosting service with CDN caching
Backend/API	Container	Spring Boot REST API is stateless, so container deployment enables fast scaling and CI/CD
Database	Managed Service	MySQL managed database gives automated backup, patching and multi-zone availability
File storage	Managed Service	Cloud object storage for resumes/reports needs durability and scaling that VM disk does not offer
AI/Python module	VM (dedicated)	NLP inference needs a stable, heavier runtime with more RAM/CPU, easiest to manage on its own VM initially
Analytics/dashboard	Container	Dashboard/reporting service is stateless and can scale independently behind the load balancer
Notification/background jobs	Serverless function	Sending emails/notifications are short-lived, event-driven tasks well suited to serverless functions

Activity 11: Final Recommendation

Final infrastructure recommendation for SkillHire:

1. Deploy on a two/three-VM production topology: an application VM (2 vCPU, 8 GB RAM) running Spring Boot + Nginx, and a dedicated NLP VM (4 vCPU, 16 GB RAM) for resume parsing and skill-gap scoring.
2. Use cloud-managed virtualization (e.g., Azure/AWS VM instances) so the hypervisor is abstracted away from the student team.

3. Containerize the backend REST API and NLP microservice with Docker for portability and CI/CD, while keeping the managed MySQL database and object storage outside the container layer.
4. Provision a managed MySQL 8.0 database (20 GB initial, multi-zone) sized independently of the app VM's RAM, growing at ~10 GB/year.
5. Allocate ~250 GB block storage for OS/application/logs and a separate ~150 GB object storage bucket for resume files, reports and images.
6. Take a VM snapshot before every major deployment, schema change, OS update and demo/review, retained for 3-7 days as a rollback safety net.
7. Use VM clones to spin up isolated QA/testing environments and team demo copies without touching the production instance.
8. Front the application with a load balancer and enable auto scaling so additional app/NLP VMs are added automatically as concurrent users grow beyond ~250.
9. Enable HTTPS, JWT authentication and daily backups across all tiers to meet the 99.9% availability target.
10. As usage scales toward 50,000+ users, move backend and NLP services fully to container orchestration (e.g., Kubernetes) with database read-replicas for horizontal scaling.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.
- Repository link submitted in Google Classroom.